



Șerban Petrescu

SVP of Software Engineering | Cloud Architect

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Professional Summary

Builder and technical leader serving as SVP of Software Engineering at Trilogy, leading 0→1 product development across a portfolio of educational technology products. Edtech work spans three pillars: learning applications (7+ apps including AI tutoring, spaced repetition games, adaptive platforms), content generation infrastructure (curriculum modeling, AI-powered authoring, publishing workflows), and learning analytics (engagement tracking, mastery measurement, outcome correlation).

Implemented learning science principles directly: spaced repetition, mastery-based progression, automaticity building, knowledge graph-driven curricula. This enables thousands of Alpha School students to achieve 99th percentile outcomes.

Player-coach approach: combine strategic leadership (\$6M budget, team of 40) with hands-on execution as product manager, architect, and full-stack engineer. Prior: 6+ years Lead Software Engineer at msg systems, SAP Developer Hero.

Key Achievements

- Shipped 7+ learning products from concept to production spanning learning apps (TrashCat, Athena, TeachTap, TeachTales), content infrastructure (Incept), and analytics (TimeBack), implementing spaced repetition, mastery-based progression, and knowledge graph-driven curricula.
- Extended Crossover hiring platform processing hundreds of thousands of monthly visitors and tens of thousands of applications, implementing AI-powered screening and assessment grading.
- Led strategic AI adoption across organization during 2024, achieving productivity gains while industry-wide metrics declined.
- Reduced cloud infrastructure costs by 50% and outages by 5x for \$20M ARR SaaS product (Playbooks).

Work Experience

SVP of Software Engineering @ Trilogy (New Products)

Feb 2023 - Present

Built independent department from 6 to 40 people while scaling from 3 to 11 concurrent products, achieving 3x productivity improvement. Managing \$6M/year budget with full autonomy over team structure and processes.

Education technology focus: Shipped 7+ learning products from concept to production spanning three pillars: learning apps (TrashCat, Athena, TeachTap, TeachTales, Magic Academy, ReadingCritters), content infrastructure (Incept platform for curriculum modeling, AI-powered content generation, QTI standardization), and analytics (TimeBack for student engagement tracking and outcome measurement).

Implemented learning science principles directly: spaced repetition, mastery-based progression, automaticity building, knowledge graphs for prerequisite relationships. This work enables thousands of Alpha School students to achieve 99th percentile outcomes. Also extended Crossover hiring platform and Trilogy Sales Platform.

VP of Software Engineering @ Trilogy (Central Engineering)

Aug 2021 - Feb 2023

Led team of 4 to 14 engineers (grown through internal transfers) operating and extending two acquired SaaS products: InsideSales Playbooks (\$20M/year ARR sales engagement platform) and ResponseTek. Focus on operational excellence and large-scale modernization of existing products.

Reduced outages by 5x and cut cloud infrastructure costs in half. Led major modernization: telephony system migration from self-hosted to serverless (Twilio), cleared defect backlog, implemented new features while maintaining production stability.

Additional Roles

Software Engineering Consultant @ msg-systems (Part-time, 1-2 hours/week)	Jun 2020 - Present
Cloud Solutions Architect @ DevFactory	Jun 2020 - Jul 2021
Teaching Assistant @ T.U. Cluj-Napoca	Feb 2019 - Jun 2019
Co-Founder @ KickIT	Sep 2018 - Jun 2020
Lead Software Engineer @ msg-systems	Apr 2014 - Jun 2020

Skills

Technical Leadership

Architecture, Code Review, Technical Decision-Making, Team Mentoring, Process Design

Management

Hiring, Performance Reviews, Budget Management, Strategic Planning, Stakeholder Management

Technologies

TypeScript, Java, NodeJS, React, Spring Boot, Unity, AWS (Lambda, AppSync, DynamoDB, RDS), GraphQL, Salesforce

Specializations

Cloud-Native Architectures, AI/ML Integration, Serverless, DevOps, Cost Optimization

Certifications

AWS Certified Solutions Architect - Professional

Jan 2021

Amazon Web Services

AWS Certified Solutions Architect - Associate

Jan 2020

Amazon Web Services

Cambridge Certificate in Advanced English (C1)

May 2011

University of Cambridge

Education

Babes-Bolyai University

Oct 2017 - Jun 2019

Master's degree in Software Engineering at the faculty of Mathematics and Informatics (dissertation pending).

Technical University of Cluj-Napoca

Sep 2012 - Jun 2016

Bachelor's degree in Computer Science at the faculty of Computer Science and Information Technology.